

EE 2080: Random Processes

Prog Assignment: Sample Questions on DTMC

Problem 1: Weather Prediction Model

Consider a simple weather model with three states:

Sunny (S), Cloudy (C), Rainy (R).

The transition probability matrix is given by:

$$P = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}.$$

1. **Generate a sample path.** Simulate and plot the weather over $n = 365$ days, starting from an initial distribution (e.g., start with Sunny on Day 1, i.e., $\pi^{(1)} = [1 \ 0 \ 0]$).
2. **Empirical long term distribution.** Starting with the initial distribution $\pi^{(1)} = [1 \ 0 \ 0]$, estimate the empirical long-term fraction of Sunny, Cloudy, and Rainy days based on your simulation. (Generate 10^6 sample paths of $n = 25$ days and compute the empirical relative frequency of the various states on the 25th day).
3. **Stationary Distribution.** Compute the stationary distribution π by solving

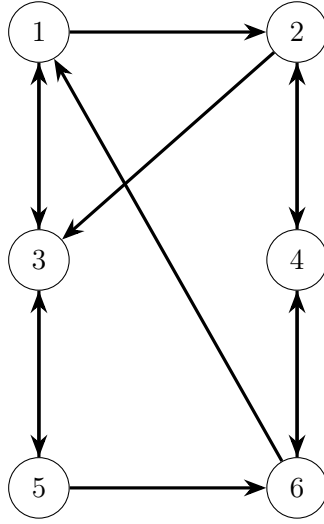
$$\pi P = \pi, \quad \sum_{i=1}^3 \pi_i = 1.$$

4. **Comparison of empirical and theoretical long term behaviour.**
Find the least number of days (n) for which the empirical distribution converges (the absolute error is $< 10^{-10}$) to the theoretical stationary distribution.
5. **Repeat for different initial distributions.** Repeat steps 2 to 4 with the initial distributions $\pi^{(1)} = [\frac{1}{3} \ \frac{1}{3} \ \frac{1}{3}]$ and $\pi^{(1)} = [0.5 \ 0.25 \ 0.25]$.

Problem 2: PageRank

Consider a web network with 6 pages represented as a directed graph. A directed edge $i \rightarrow j$ denotes a hyperlink from page i to page j .

Pages: $\{1, 2, 3, 4, 5, 6\}$



1. **Construct the transition matrix P .** For each node i , let $\text{outdeg}(i)$ be its number of outgoing edges. Define

$$P_{ij} = \begin{cases} \frac{1}{\text{outdeg}(i)}, & \text{if there is an edge } i \rightarrow j, \\ 0, & \text{otherwise.} \end{cases}$$

If a node has no outlinks (*dangling node*), set its entire row to the uniform distribution $1/n$ (here $n = 6$).

2. **Form the Google matrix.** With damping factor $\alpha = 0.85$, define

$$G = \alpha P + (1 - \alpha) \frac{1}{n} \mathbf{J}_n, \quad n = 6,$$

where $\mathbf{J}_n$ is the $n \times n$ all-ones matrix.

3. **Compute the PageRank vector π .** Implement *both* methods below and compare:

- i. **Power iteration .** Initialize $\pi^{(0)} = \frac{1}{n} \mathbf{1}$. Iterate

$$\pi^{(k+1)} = \pi^{(k)} G,$$

until convergence, e.g. $\|\pi^{(k+1)} - \pi^{(k)}\|_1 < 10^{-10}$ with a cap $k_{\max} = 10000$. Normalize $\pi^{(k)}$ to sum to 1 if needed.

- ii. **Linear system.** Solve the stationary equation $\pi G = \pi$, $\sum_i \pi_i = 1$.

Problem 3: Hitting Times and Mean Recurrence Times

Consider a time homogeneous discrete-time Markov chain with state space

$$S = \{1, 2, 3, 4\}$$

and the transition probability matrix

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.2 & 0.5 & 0.3 & 0 \\ 0 & 0.3 & 0.4 & 0.3 \\ 0.1 & 0 & 0.4 & 0.5 \end{bmatrix}.$$

1. **Hitting time simulation.** Let $T_{i \rightarrow A}$ denote the first hitting time of a set A starting from state i . Here, take $A = \{4\}$ and write code to simulate the chain starting at state 1 to estimate

$$h_{1,4} \triangleq \mathbb{E}[T_{1 \rightarrow \{4\}}].$$

Run at least 10^5 independent sample paths and report the empirical mean.

2. **Empirical mean recurrence time.** Run a single long sample path, (a.k.a trajectory) (e.g., 10^6 steps) of the chain and estimate the average return time to state 2:

$$\hat{m}_2 = \frac{1}{K} \sum_{k=1}^K (\tau_k - \tau_{k-1}),$$

where $0 = \tau_0 < \tau_1 < \dots < \tau_K$ are successive visit times to state 2.

3. **Stationary distribution and mean recurrence times.** Compute the stationary distribution π by solving

$$\pi P = \pi, \quad \sum_{i \in S} \pi_i = 1, \quad \pi_i \geq 0.$$

Use the fact that for an irreducible positive recurrent DTMC, the mean recurrence time of state i is

$$m_i = \frac{1}{\pi_i}.$$

Report m_2 from π and compare $\hat{m}_2$ with $m_2 = \frac{1}{\pi_2}$.

Problem 4: Communicating Classes and Class Properties

Consider a time homogeneous discrete-time Markov chain with state space

$$S = \{1, 2, 3, 4, 5, 6\}$$

and the transition probability matrix

$$P = \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0.2 & 0 & 0.8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 0.3 & 0.7 & 0 \end{bmatrix}.$$

1. **Communicating Classes:** Identify the communicating classes.
 - Generate $N = 500$ independent trajectories till time $T = 10$.
 - Create an empirical $|S| \times |S|$ reachability matrix $\mathbf{R}$ with $\mathbf{R}_{i,j} = 1$, if there exist at least one trajectory in which there is a path from i to j , else $\mathbf{R}_{i,j} = 0$. If $\mathbf{R}_{i,j} = 1$ and $\mathbf{R}_{j,i} = 1$, i and j are in the same communicating class.
2. **Empirical verification of transience or recurrence** Empirically classify the states of the DTMC as transient or recurrent.
 - For each state you want to classify, set $X_0 = i$ and run $N = 500$ independent trajectories till time $T = 100$.

- For each run r , record the time till first return to state i , i.e., $t_r^i = \min(n \geq 1 : X_n = i)$, if $t_r \leq T$ and an event indicator $\delta_r = \mathbb{I}_{t_r \leq T}$.
 - Compute Relative frequency of ever-returning to state i within T , i.e., $f_{i,i}(T) = \frac{1}{N} \sum_{r=1}^N \delta_r$
 - If $f_{i,i}(T) < 1$, label the state **transient**, else **recurrent**.
3. **Verification of class properties** For each of the communicating classes in step 1), verify that the states within a class are all transient or all recurrent, i.e., transience and recurrence are class properties.